WEB SYSTEMS AND TECHNOLOGIES

FINAL PROJECT PROPOSAL

Uniformity: An E-Commerce platform for Pre-Loved School and Professional Uniforms

GROUP NAME: TechNic **Course & Block:** BSIT-2A		
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Submitted to:

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Assistant Professor IV

Title: Uniformity: An E-Commerce platform for Pre-Loved School and Professional Uniforms

Introduction:

Uniformity is a web-based marketplace designed to help users buy and sell pre-loved uniforms safely and conveniently. Many students and professionals have uniforms that are still in good condition but no longer in use. At the same time, others may struggle to find affordable uniforms in the right sizes. This project will provide a centralized web platform where users can list, browse, and purchase uniforms efficiently. The platform supports both buying and selling within a single account, offering a simple and user-friendly experience.

System Purpose:

The purpose of Uniformity is to provide an organized, secure, and accessible system for exchanging uniforms. It promotes sustainability through the reuse of uniforms while offering users a convenient way to manage buying, selling, and order tracking. The system also integrates product management, cart functionality, and a checkout process to simulate a real-world online marketplace experience.

Target Users:

- Students looking for affordable uniforms
- Professionals in need of workplace uniforms
- Users who want to sell unused or extra uniforms

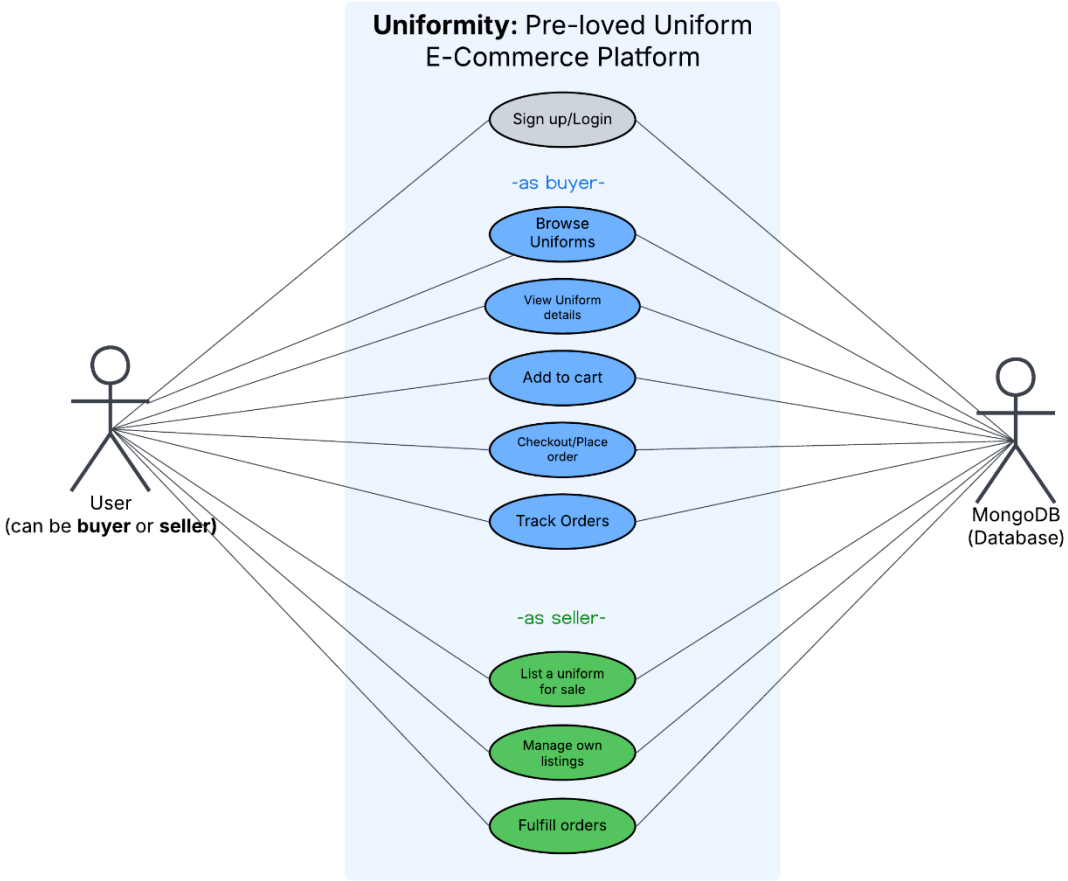
Key Features:

- **User Authentication –** Secure registration and login with JWT
- **Product Listing (Seller Feature) –** Upload uniforms with details such as name, size, quantity, price, and image
- **Product Browsing & Viewing –** Browse uniforms, view details, and see seller profile information
- **Cart & Checkout System –** Add items to a cart, provide delivery information, and place orders via a mock payment system
- **Order Management –** Track orders in user profile

Objectives:

- Develop a fully functional web-based marketplace for uniforms
- Simplify buying and selling processes for users
- Promote sustainability by reusing uniforms
- Provide structured order and product management features
- Deliver a responsive and user-friendly interface

I. Use Case Diagram (UML)



Sign up / log in — users create an account or log in using JWT-secured authentication to access the platform.

Browse uniforms — users can view all available uniform listings filtered by category, size, or price.

View uniform details — users can see full product info including images, size, price, and seller profile.

Add to cart — users can add desired uniforms to their cart before purchasing.

Checkout / place order — users provide delivery address and complete the order via mock payment.

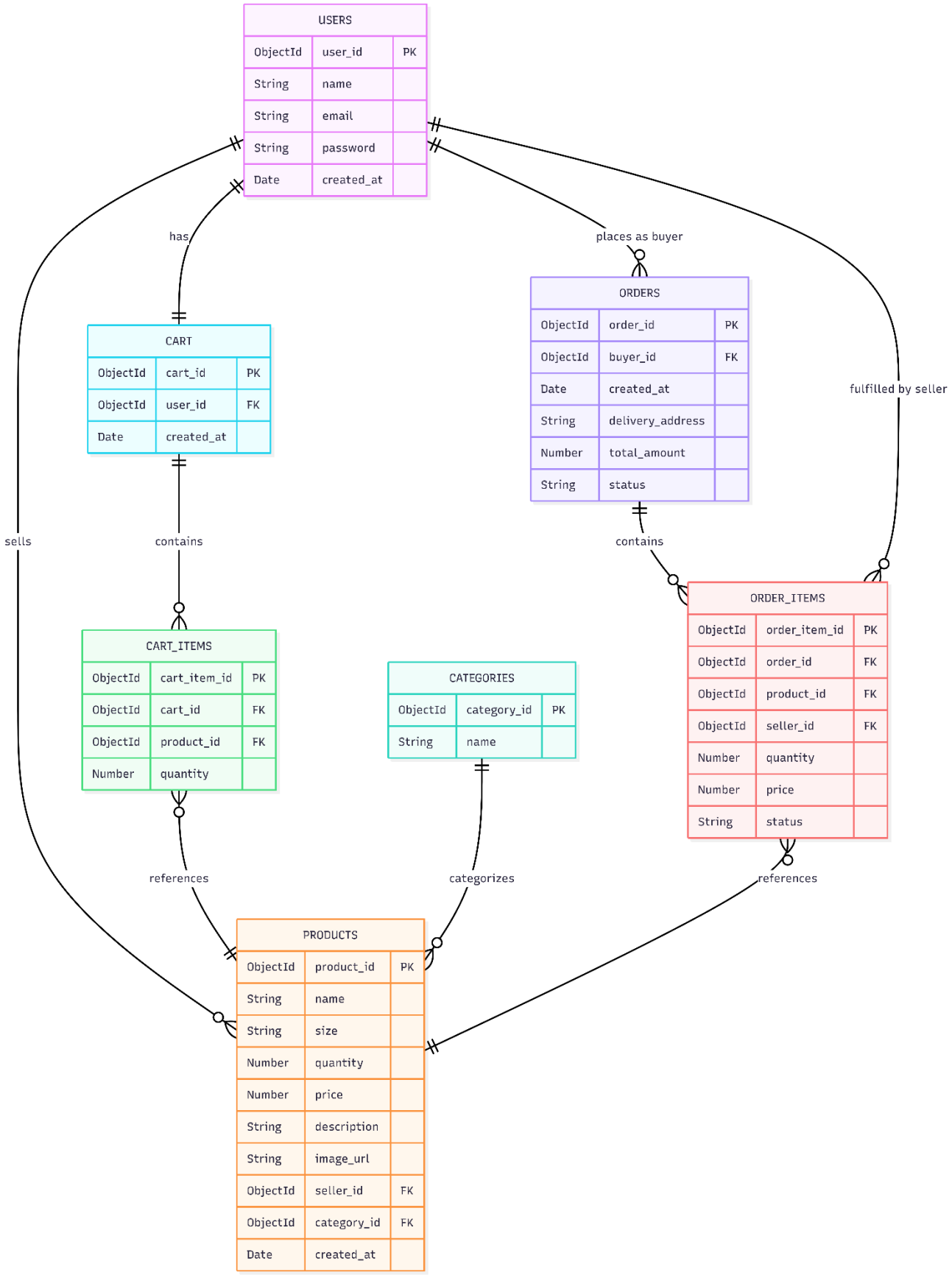
Track orders — users can monitor the status of their placed orders from their profile.

List a uniform for sale — sellers can upload a uniform with name, size, quantity, price, description, and image via Cloudinary API.

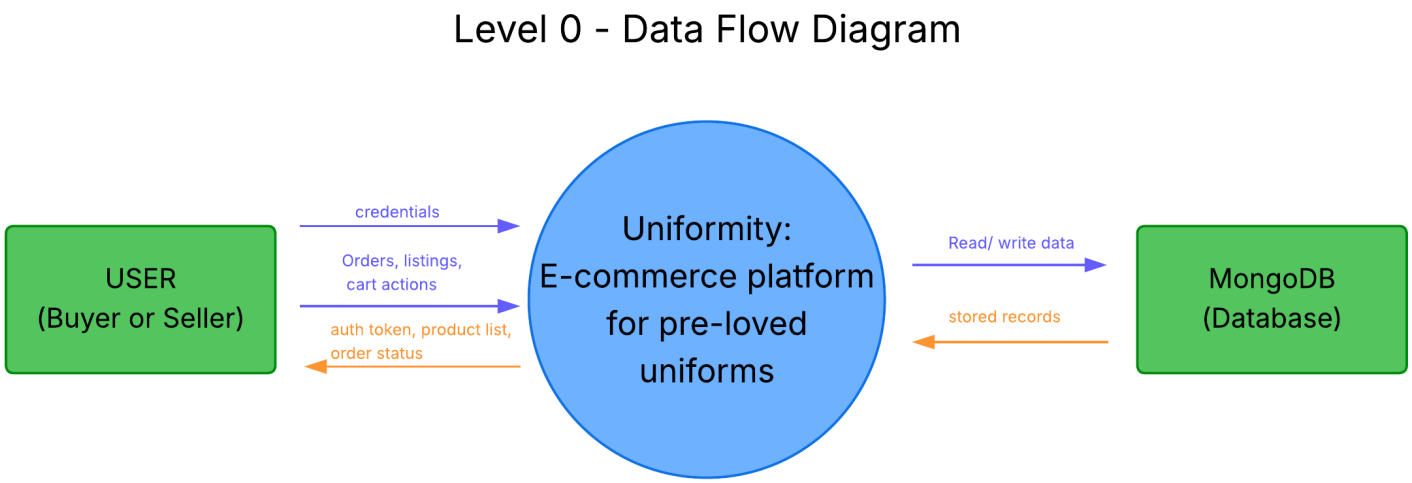
Manage own listings — sellers can edit or remove their listed uniforms.

Fulfill order — sellers can update the status of orders placed for their listed uniforms.

II. Entity Relationship Diagram (ERD)



III. Data Flow Diagram (Level 0)



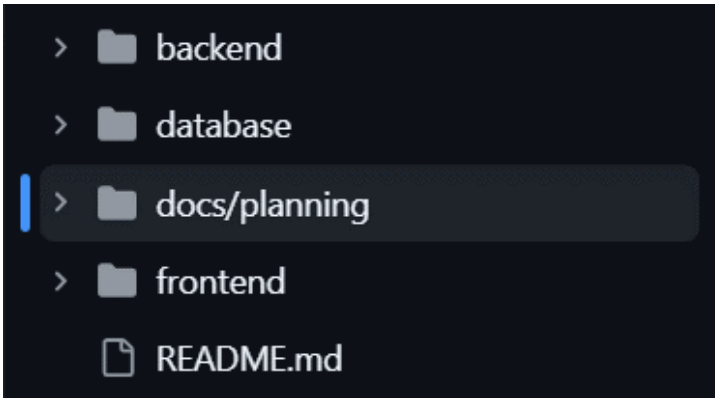
This Level 0 DFD shows the overall picture of our system.

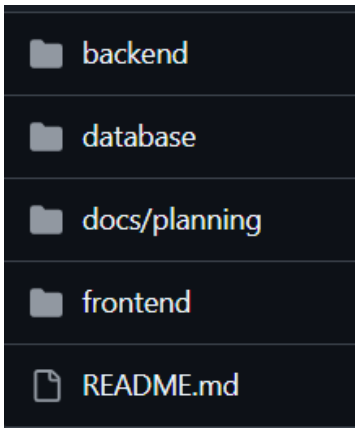
- User — the person interacting with the system, either as a buyer or seller. They send in things like login credentials, orders, cart actions, and product listings. In return, they receive their auth token, product lists, and order status updates.
- Uniformity — the entire system represented as one single process. It receives inputs from the user, processes them, and communicates with the database.
- MongoDB — the external database that the system reads from and writes to. All data such as users, products, orders, and categories are stored and retrieved here.

IV. GitHub Repository URL

Link: https://github.com/red5458/final-project-TechNic-BSIT2a

V. Screenshot of Repo Structure





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